Phase 1 Report: Nepali (Model L)

Swaraj Kumar
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Abstract

This is a standalone Phase 1 report for **Nepali as Model L**, companion to `phase1_report.tex` (Hindi / Bhojpuri). It follows the same structure but is a separate document because Nepali runs on a different pipeline (`nepali/scripts/`, stage-based S00–S14) with a different data layout than the Hindi/Bhojpuri cleaner, so its own numbers are not simply insertable into that report without changing what each figure measures. Nepali's final corpus is **635,204,333 tokens** (127.0% of the ∼500M planning target), clearing both of the pipeline's absolute lane targets (manual ≥100M, downloaded ≥400M) and the assignment's 20% manual-collection floor (26.41%, token-weighted). Nepali supersedes Bhojpuri as the graded Model L; Bhojpuri's own work is kept intact in the repository and in `phase1_report.tex` as a complete, working record, not deleted.

Contents

1 Overview **2**

2 Data Collection **2**

3 Cleaning Pipeline **3**

4 Splits **5**

5 Tokenizer **8**

6 Cross-Language Check (Nepali vs. Hindi) **11**

7 Reproducibility **11**

8 Limitations **14**

1 Overview

Model	L (lower-resource)
Language	Nepali (allowed-list language, course spec §1.1)
Final corpus size	635,204,333 tokens (127.0% of the ∼500M planning target)
Manual share	26.41% of final tokens (floor is 20%)
Tokenizer	BPE, vocab_size = 10,000
Train / val / test docs	1,520,000 / 199,428 / 198,837

Table 1: Nepali corpus at a glance.

Nepali's pipeline enforces two **absolute** token targets rather than a single overall total: manual lane ≥ 100,000,000 tokens, downloaded lane ≥ 400,000,000 tokens (`nepali/configs/language.yaml`). Both are met:

Lane	Tokens	Target	Met
Manual	167,787,251	≥100,000,000	Yes
Downloaded	467,417,082	≥400,000,000	Yes
Total	**635,204,333**	∼500,000,000 (course target)	127.0%

Table 2: Absolute lane targets, both met. Source: `nepali/data/final/corpus_stats.json`.

2 Data Collection

Manual vs. downloaded, as defined for this pipeline

Same course ruling as Hindi/Bhojpuri: "manual" means content requiring active per-source collection work (API/XML-dump parsing, HTML scraping with site-specific extraction, OCR) as opposed to downloading an already-corpus-ified public dataset. Unlike the Hindi/Bhojpuri report's char-weighted manual share, Nepali's split is **token-weighted** – the pipeline tracks words/tokens per lane past S03, not characters. This is a methodology difference worth being explicit about if the two reports are ever compared directly.

Manual-lane sources

Downloaded-lane sources

Why the manual lane is dominated by one source

`news_onlinekhabar` alone supplies 91.6% of manual tokens. This is a real, measured constraint, not a collection shortcut: OnlineKhabar has the largest scrapable Nepali news archive among the manual sources configured, and `nepali/scripts/auto_grow_and_tokenize.py`'s growth loop bumped its `target_words` in `sources.yaml` (capped at 260,000,000 words) specifically because it was the only manual source with enough real headroom to close the gap to the 100M-token manual floor. The other manual sources (Wikipedia, Wiktionary, Wikibooks, smaller news sites, one OCR document) are genuinely smaller pools for Nepali than their Hindi equivalents.

Source	Docs	Words	Tokens
news_onlinekhabar (web scrape)	308,139	107,614,170	153,717,439
wikipedia_api (XML dump, own parser)	16,445	4,622,944	8,052,774
news_bbc_nepali (web scrape)	12,377	3,555,706	5,277,060
wikibooks_api	74	230,885	412,110
wiktionary_api	606	96,765	200,300
news_ekantipur	119	80,432	115,316
news_ratopati	20	6,055	10,608
ocr_archive_books	1	447	866
news_nagarik	2	503	778
Manual total	**337,783**	**116,207,907**	**167,787,251**

Table 3: Manual-lane sources.

Source	Docs	Words	Tokens
sangraha_verified (ai4bharat/sangraha)	579,743	173,902,717	258,462,424
sangraha_unverified (ai4bharat/sangraha)	341,901	99,767,699	141,886,279
indiccorp_v2 (ai4bharat/IndicCorpV2)	658,838	45,407,223	67,068,379
Downloaded total	**1,580,482**	**319,077,639**	**467,417,082**

Table 4: Downloaded-lane sources. Source for both tables: `corpus_stats.json`, `by_source` breakdowns summed across train/val/test.

3 Cleaning Pipeline

Nepali's pipeline is stage-based (`nepali/scripts/s00_doctor.py` through `s14_corpus_stats.py`), sharing the same `corpuslib` code as Hindi's own Nepali-parallel pipeline copy (byte-identical, hashed at CI time) so the two pipelines cannot silently drift apart while still operating on fully separate data.

Document retention through the pipeline

Stage	Docs in	Docs out	Retention
S03 normalise	2,655,504	2,655,504	100.00%
S04 language ID	2,655,504	2,649,643	99.78%
S07 train-pool split-off	2,649,643	2,251,378	84.97%
S05 heuristic filters	2,251,378	2,245,354	99.73%
S06 perplexity filter	2,245,354	2,178,418	97.02%
S08 dedup (exact + near)	2,178,418	2,126,334	97.61%
S09 decontaminate (vs. val/test)	2,126,334	2,109,223	99.20%
S10 balance lanes (train only)	2,109,223	1,520,000	72.06%

Table 5: Document retention through the Nepali pipeline.

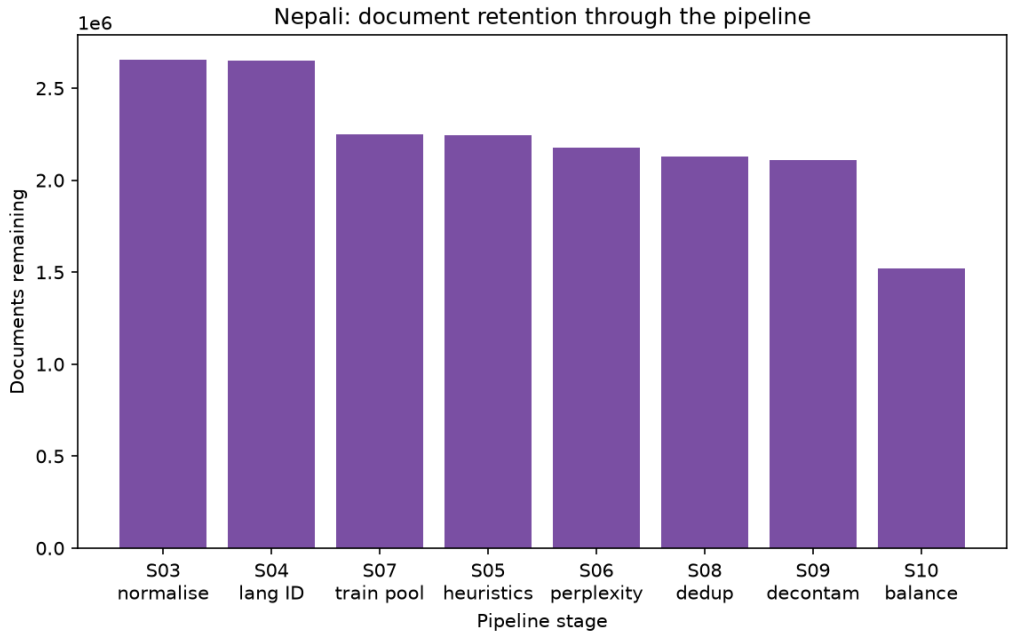


Figure 1: Document retention through the pipeline.

S04 language-ID rejects (largest reasons)

Reason	Docs
wrong_language	1,599
low_confidence	1,189
narrow_margin_vs_hi	600
too_short	251
other script-ratio buckets ($\leq$5 docs each)	121

Table 6: S04 language-ID reject reasons.

`narrow_margin_vs_hi` (600 docs) is notable given Nepali and Hindi share the Devanagari script and overlapping vocabulary – this is the language-ID model catching documents too ambiguous to confidently call Nepali rather than Hindi, which is exactly the failure mode the cross-language check in Section 6 independently verifies did not leak through.

S05 heuristic-filter drops

S08 deduplication

Removed 52,084 of 2,178,418 documents: 2,350 exact duplicates, 49,734 near duplicates (MinHash/LSH, `num_perm=100, bands=20, rows=5, shingle_size=5, threshold=0.55`). 21,273 cross-lane duplicate pairs were found (the same content collected via both the manual and downloaded lanes); all 21,273 were resolved in favor of the manual copy, per the pipeline's priority-to-manual policy.

Reason	Docs
repeated_lines	1,858
too_few_sentences	1,226
too_short	382
link_farm	481
low_lexical_diversity	5
too_few_words	8
mean_word_too_long	1

Table 7: S05 heuristic-filter drop reasons.

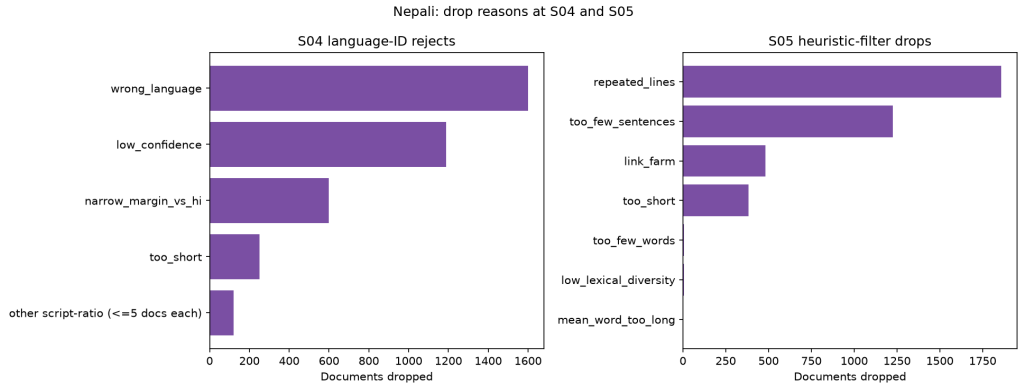


Figure 2: S04 language-ID rejects and S05 heuristic-filter drops.

S09 decontamination

Removed 17,111 train documents that matched held-out (val+test) content above threshold 0.55, preventing train/eval leakage at the content level (not just document-id level). A rescan afterward found 0 remaining matches.

S10 lane balancing

The train pool's raw manual share going into S10 was 23.27%. Rather than loosening any filter to inflate it, S10 trimmed the **downloaded** lane (589,223 documents dropped, manual lane untouched) to bring the ratio to 27.42% – safely above the 20% floor without manufacturing volume.

4 Splits

Val and test were held out at S07, *before* S10's lane balancing – which is why their manual share (23–24%) differs slightly from train's post-balancing 27.42%; both are comfortably above the 20% floor regardless. Zero document-id overlap between all three splits, confirmed by `nepali/data/final/split_stats.json` (`zero_doc_id_overlap: true`, `flagged_splits: []`) and independently re-checked by `scripts/validate_phase1_nepali.py`.

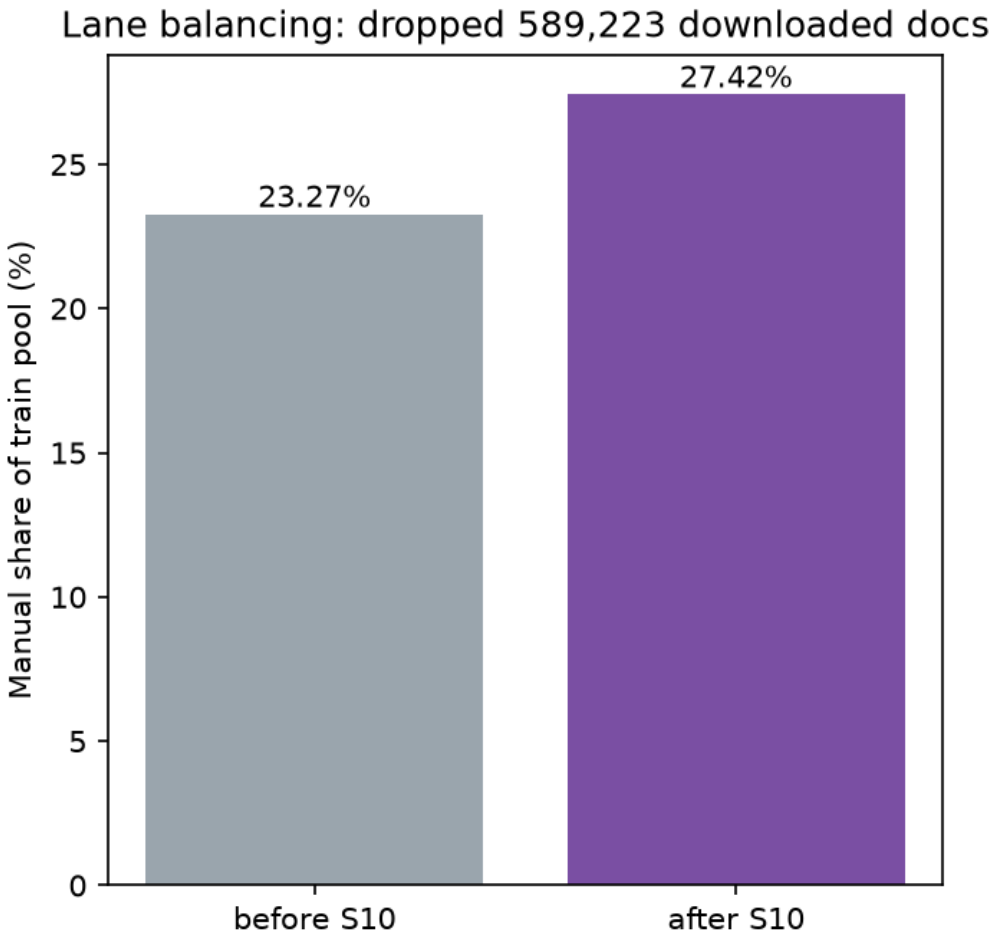


Figure 3: Lane balancing: manual share before/after S10.

Split	Docs	Words	Tokens	Manual share
Train	1,520,000	356,691,169	519,906,359	27.42%
Val	199,428	39,432,622	57,839,396	23.71%
Test	198,837	39,161,755	57,458,578	23.11%
Overall	**1,918,265**	**435,285,546**	**635,204,333**	26.70% (word) / 26.41% (token)

Table 8: Final splits.

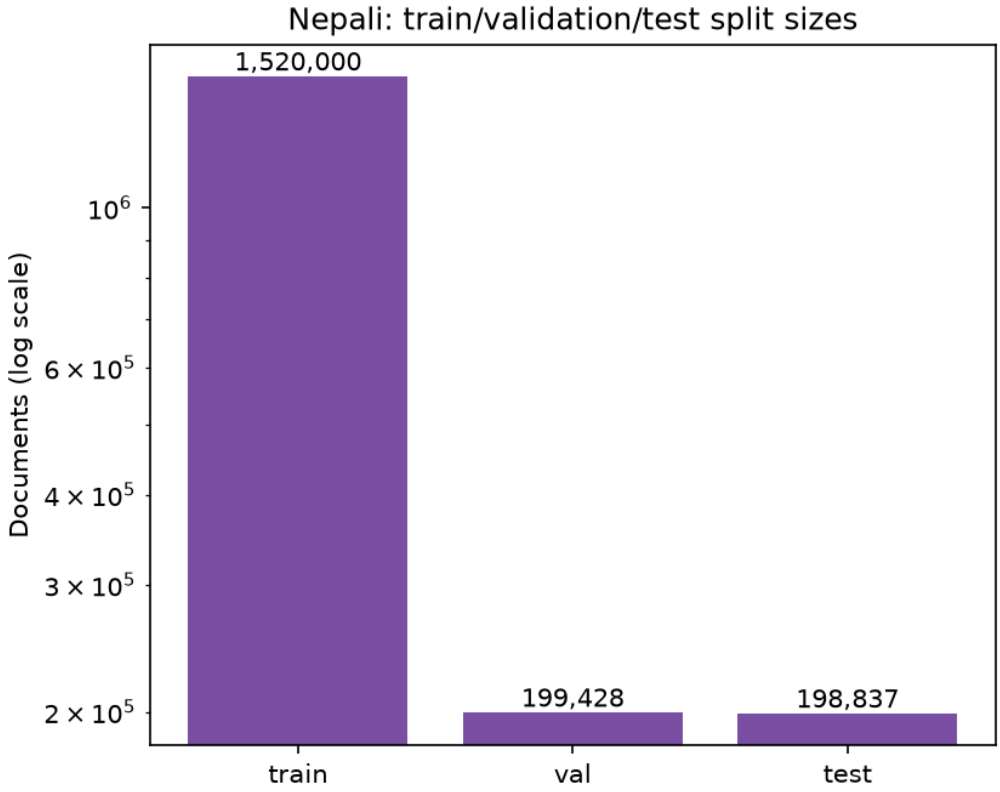


Figure 4: Train/validation/test split sizes.

5 Tokenizer

Algorithm and sweep

BPE (SentencePiece), swept at vocab_sizes [5000, 6000, 8000, 10000] per `nepali/configs/tokenizer.yaml`, matching the course-TA ruling capping recommended vocab size to the 5,000–10,000 range given the ∼25M-parameter model budget (the same ruling applied to Hindi's own vocab_size=8000 choice).

Vocab size	Fertility (tok/word)	Chars/token	UNK rate
5,000	1.7032	3.7976	0.0%
6,000	1.6378	3.9495	0.0%
8,000	1.5472	4.1808	0.0%
10,000 (chosen)	**1.4868**	**4.3505**	**0.0%**

Table 9: Tokenizer vocab sweep.

Selection note (from `nepali/tokenizer/selection_report.json`): "no elbow found within the swept range; fertility was still improving by more than 3% at every step, so the largest candidate (10,000) was selected" – i.e. the sweep's own stopping rule, not a manually picked number.

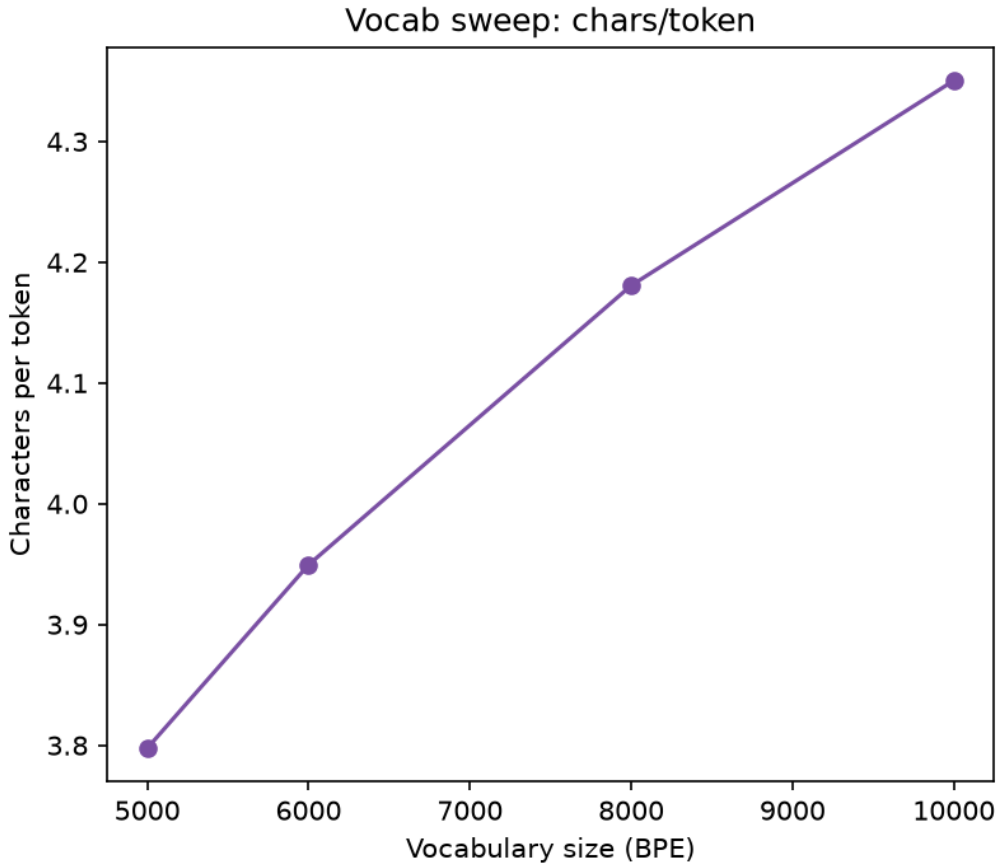


Figure 5: Vocab sweep: characters per token.

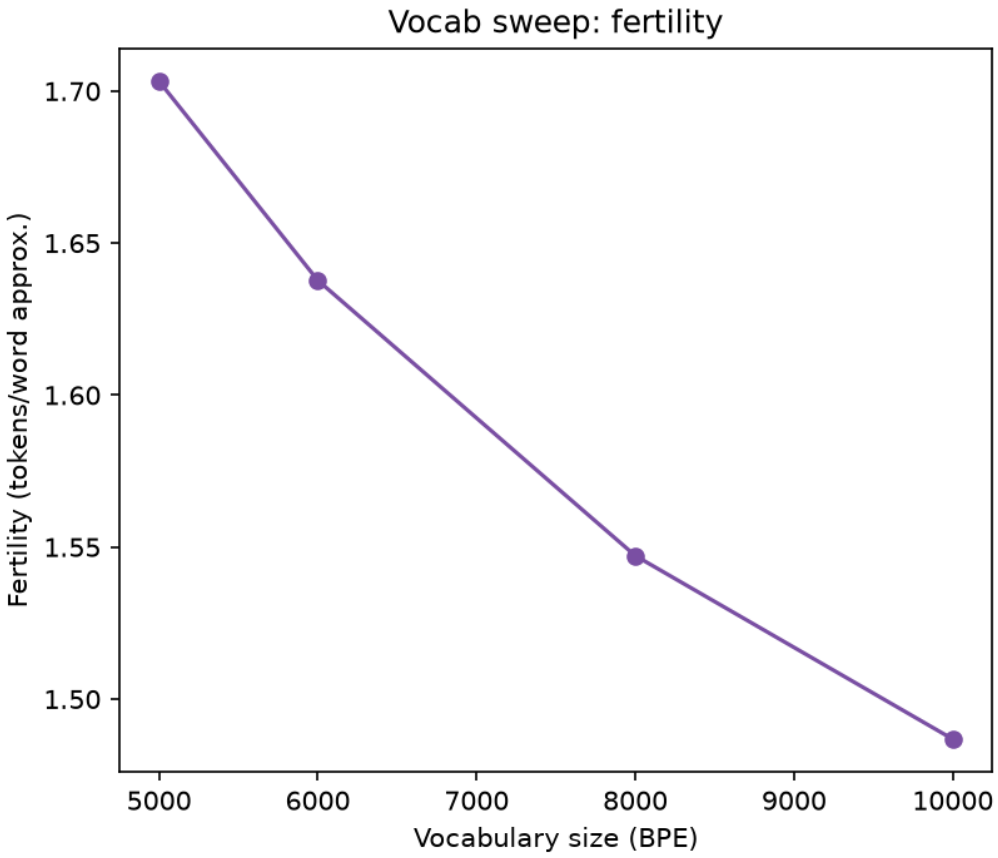


Figure 6: Vocab sweep: fertility.

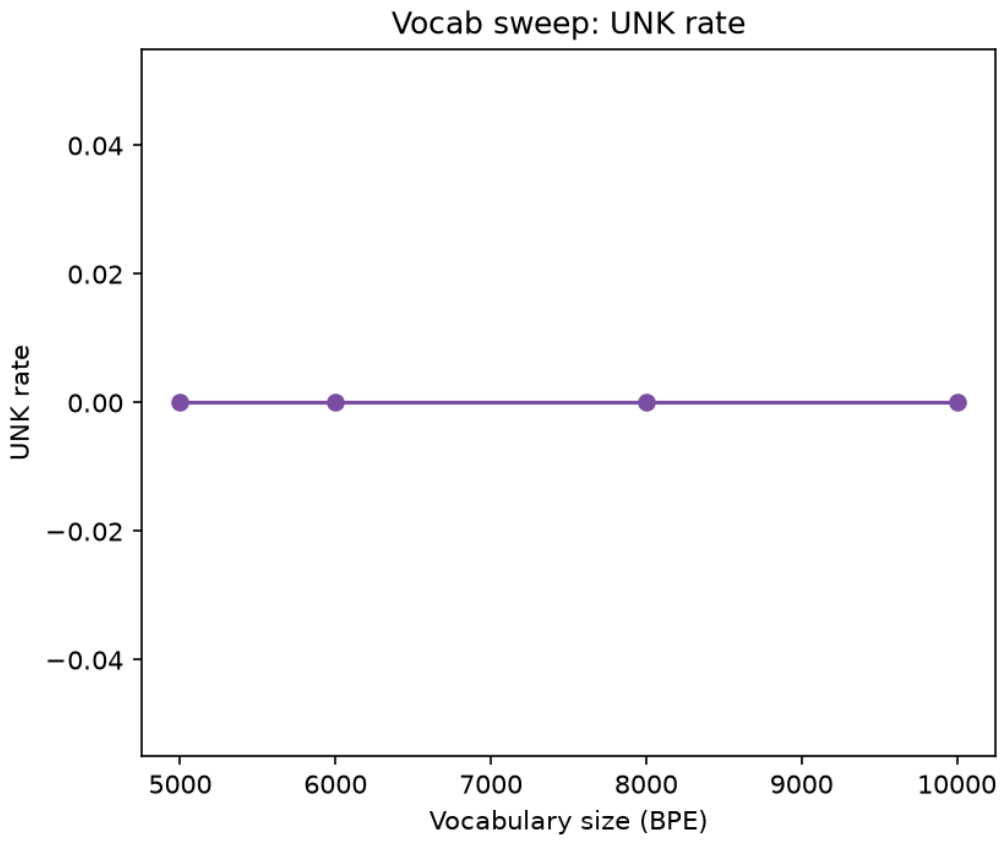


Figure 7: Vocab sweep: UNK rate.

Held-out test-split evaluation (chosen model, vocab_size=10,000)

Metric	Value
Test fertility	1.4672 tokens/word
Test chars/token	4.405
Test UNK rate	0.0%
Train chars/token	4.439

Table 10: Held-out test-split evaluation. Source: `nepali/tokenizer/evaluation_report.json`.

Sample tokenizations (real model output, vocab_size=10,000)

नेपाली पद वर्ग सङ्खेय नाम अर्थ कमजोर भएको अवस्था; गल्ती → _नेपाली _पद _वर्ग _सङ् खे य _नाम _अर्थ _कमजोर _भएको _अवस्था ; _गल्ती ... (45 tokens total)

नेपाली पद वर्ग नाम अर्थ मान्छे; मानसि → _नेपाली _पद _वर्ग _नाम _अर्थ _मान्छे ; _मानसि ... (45 tokens total)

Full examples with complete piece sequences are in `nepali/tokenizer/evaluation_report.json`.

Top tokens by frequency (top 50, training corpus)

Most frequent pieces are punctuation and function morphemes, as expected for a subword tokenizer on inflected Devanagari text: _। (16.1M), , (9.6M), _र (“and”, 6.7M), _छ (copula, 6.7M), । (5.3M), को/मा/का/ले (case/postposition suffixes, 3.2–4.7M each).

6 Cross-Language Check (Nepali vs. Hindi)

Run via `scripts/check_nepali_hindi_overlap.py`: every one of Nepali’s 1,918,265 final documents (train 1,520,000 + val 199,428 + test 198,837) had its `dedup_key` (sha1 of whitespace-collapsed, punctuation-stripped, lowercased text – identical algorithm to `scripts/clean_lib.py:dedup_key`, computed fresh since Nepali’s own pipeline doesn’t store one) checked against the full set of 704,101 dedup_keys already computed for Hindi’s cleaned corpus (`hindi/data/cleaned/metadata.jsonl`).

Result: PASS – 0 shared dedup_keys. No document is shared between the Nepali and Hindi corpora. Full run took 467.7 seconds. This addresses the spec’s “do not share documents across corpora” requirement for the two languages that are actually part of the graded submission (Nepali vs. Bhojpuri was not checked, since Bhojpuri is superseded and not part of the final two-language submission).

7 Reproducibility

```
cd nepali/scripts
python s00_doctor.py                    # environment check
python auto_grow_and_tokenize.py        # full S02-S14 pipeline, grows lanes to target
python ../../scripts/validate_phase1_nepali.py
python ../../report/phase1/make_plots_nepali.py
```

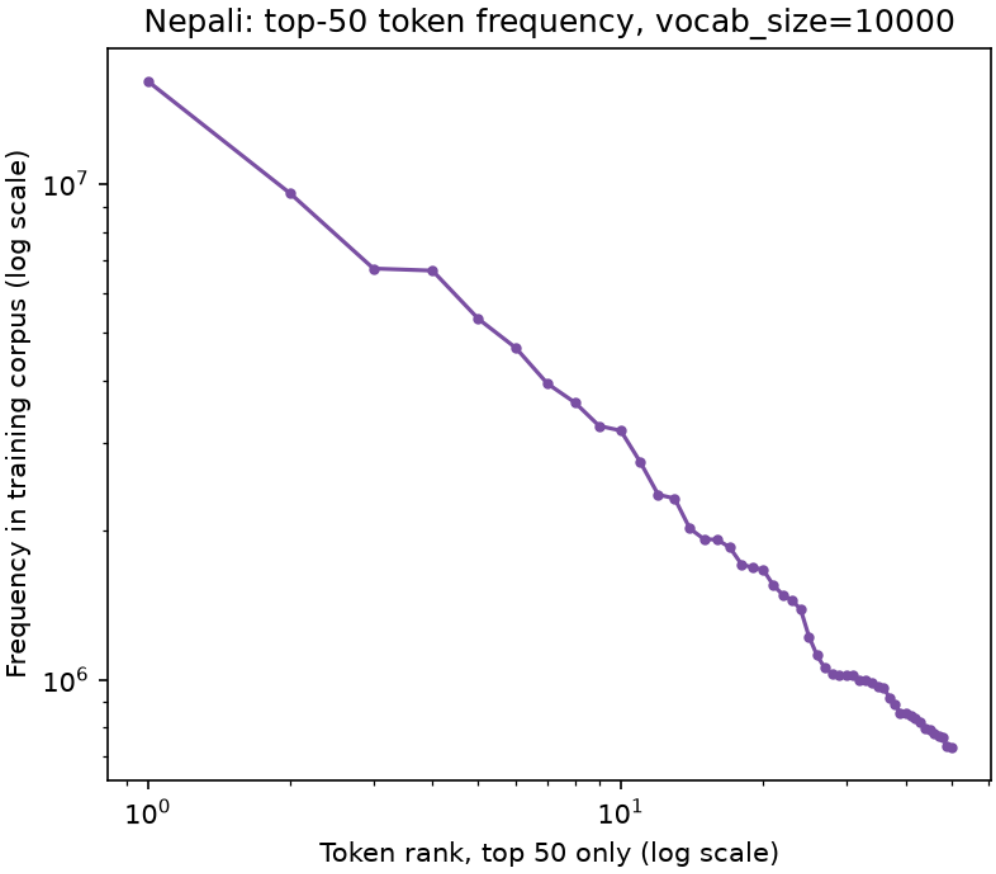


Figure 8: Top-50 token frequency (log-log), vocab_size=10,000. Labelled explicitly as top-50, not the full distribution – see caveat below.

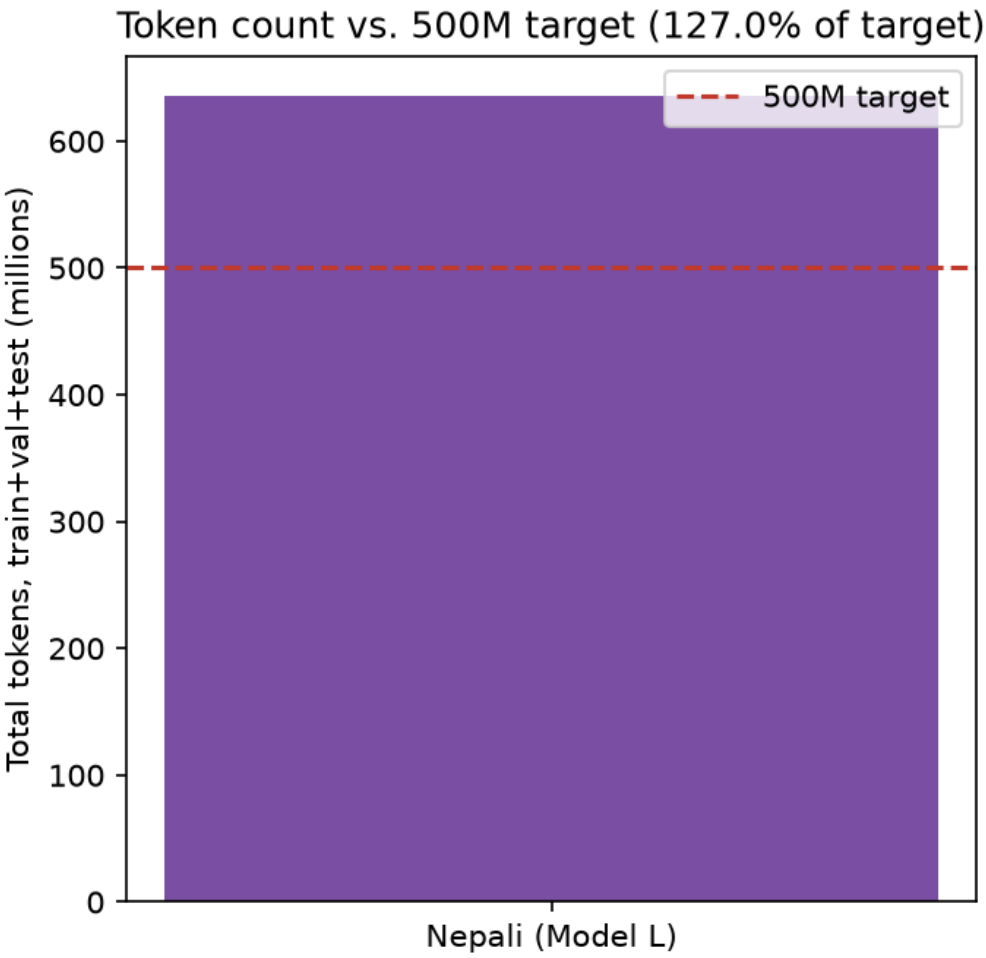


Figure 9: Total token count vs. the 500M target.

```
python ../../scripts/check_nepali_hindi_overlap.py
```

All figures and this report's numbers are direct reads of `nepali/data/final/corpus_stats.json`, `split_stats.json`, `nepali/tokenizer/selection_report.json`, `evaluation_report.json` – nothing here is hand-transcribed or estimated.

Provenance note: the corpus/tokenizer artifacts referenced in this report were produced by running `nepali/scripts/` on a separate machine (the primary machine's disk/time constraints made the 30+ hour S08 dedup pass impractical there); only the pipeline's output files were transferred into this repo, not live per-stage console logs. `log.md`'s S08(production)–S14 entries are reconstructed from those output files rather than logged in real time – see the note inline in `log.md` for exactly which entries this applies to and their sourcing.

8 Limitations

- **Manual share is token-weighted, not char-weighted** (see §2), unlike the Hindi/Bhojpuri report. Character-level lane tracking was never added to this pipeline; token-weighted is the honest number available.
- **Cross-language check covers Nepali vs. Hindi only** (§6), not Nepali vs. Bhojpuri, since Bhojpuri is not part of the final two-language submission.
- **Top-token frequency (Fig. 8) is top-50 only**, not a full Zipf distribution, since `evaluation_report.json` only retained the top 50 and reproducing the full distribution would require re-encoding the entire corpus.
- **Manual-lane concentration**: 91.6% of manual tokens come from one source (OnlineKhabar). This is disclosed in §2 rather than smoothed over.